PRAgent: AI-Assisted Physics Paper Reproduction

Capabilities and Evidence from a 100-Paper Project

Technical report draft; author metadata pending

September 2026

Abstract

PRAgent connects physics-paper interpretation, numerical implementation and scientific assessment in a human-guided research workflow. We examine the system through public implementations and a documentary audit of a fixed 100-paper project. Three cases provide concrete evidence: resource-bound code that refines large integer decisions with 60-digit arithmetic; quantum dynamics implemented in conserved particle sectors; and finite-ensemble discord calculations that expose a discrepancy with an analytical reference. A surface-code case illustrates a system-level distinction: checks pass for an implemented repetition-code proxy, while the original paper target remains unaccepted. To evaluate the breadth of these results, we trace 3,241 archived result-object records from 95 papers to their declared comparison conditions and references, and reconcile summary and detailed assessment records. This audit finds that the 26 papers with all associated checks historically labelled reproduced and the 26 with summary-level review marked passed are disjoint. Consequently, check-label fractions cannot serve as a scientific reproduction rate. The contribution is an inspectable implementation and assessment workflow, supported by numerical examples and explicit evidence links. The archive establishes bounded capabilities within that workflow; it does not yet establish an autonomous-agent success rate or complete reproduction of the cohort.

1 Introduction

A physics paper specifies a scientific argument, but rarely a complete software task. Reproducing it requires decisions about notation, state-space representation, numerical precision, observables and comparison conditions. A program can run successfully while computing a different model or testing a weaker claim. A useful scientific agent must therefore connect implementation to the physical question and make that connection available for inspection.

PRAgent is an AI-assisted system for this work. It organizes paper interpretation, numerical implementation and assessment around a persistent research case. Its public output, distributed through RunThePaper, includes derivations, case-specific code, generated arrays and explanations of what the calculation establishes. The system is used with human guidance; throughout this report, demonstrated capability refers to what is implemented and evidenced in that combined workflow.

We make these capabilities assessable through source-target-to-evidence chains. Three detailed cases supply the numerical examples. Hamiltonian-simulation resource estimates require stable arithmetic and precision-aware integer decisions. Quantum walks use conserved particle sectors to construct exact finite-dimensional dynamics. Quantum-discord calculations distinguish finite ensembles from an analytical reference and from an unpublished author realization. A further case shows how the assessment records distinguish local model checks from acceptance of the source target. A uniform documentary audit then tests whether aggregate labels across the fixed 100-paper project can support a broader performance claim. The technical contribution is the implemented research system and its inspectable evidence across these tasks; the source papers supply the underlying physical methods.

2 Research context

Traceable computation and accessible research materials are central to computational reproducibility [1,2]. They motivate PRAgent's persistent case structure: a numerical figure should be connected to equations, inputs, implementation and checks. This also provides a way to evaluate an agent's scientific work through the objects it produces.

PaperBench, CORE-Bench and MLE-bench evaluate complementary forms of research replication and computational engineering [3–5]. The present report is a capability study of a system used across heterogeneous physics tasks. It examines implemented methods and their evidence; comparative performance under matched budgets remains an experimental question. The detailed physical examples implement

qDRIFT resource formulas, a quantum-walk model and DQC1 discord calculations from their respective source papers [6–8].

3 PRAgent capabilities and system design

3.1 Connecting the scientific task to executable code

A PRAgent research case records the paper identity, scientific scope, equations, configuration, implementation, generated outputs and assessment. A source object identifies a curve, table entry or analytical statement. An execution target specifies the computation that addresses it. Artifacts preserve the resulting arrays, diagnostics and explanation. One propagation can support several observables, while one plotted result may require several convergence checks; these relationships remain explicit in the case.

The numerical task fixes the model, observable, system size, parameters, boundary conditions and approximation regime. Equation records connect the source expressions to the functions and configuration fields that implement them. This connection is where the case-specific physics enters: a conserved sector can define an exact basis, a resource estimate can require inequality evaluation, and an ensemble study must specify which distribution is sampled.

3.2 Implementing and checking the calculation

Case implementations separate model construction, solution, observables and rendering. This allows generated numerical tables to be inspected independently of the plot, and lets several observables share one solver output. Checks are tied to the scientific object: conservation for dynamics, neighboring integer inequalities for resource bounds, and sampling intervals for an ensemble. Control calculations test the consequences of changing a physical assumption.

The workflow in Fig. 1 connects these computations to assessment and revision. PRAgent retains the target, configuration and evidence associated with each result, including unresolved comparisons. This supports targeted follow-up: revise the model, investigate a convention, extend a calculation or obtain a missing input. The scientific meaning of a check is preserved alongside its pass or fail status.

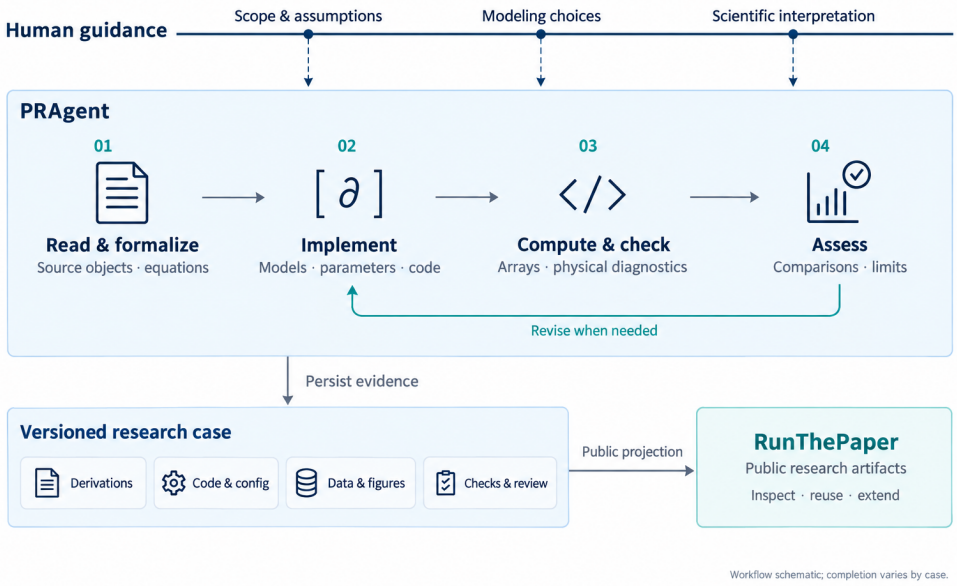


Figure 1: PRAgent's human-guided research workflow. Paper interpretation, numerical implementation and assessment produce a persistent research case. Human guidance can shape the task and revisions; case-specific code performs the numerical work. Derivations, arrays and review records remain linked to the result, and selected artifacts enter the public RunThePaper collection. Dashed connections indicate human guidance; solid arrows indicate workflow or artifact transfer.

3.3 A concrete assessment chain: a valid proxy is an incomplete target

Case `2502.20558`, target T001, makes the assessment model tangible. The source target is the logical-error-versus-rounds comparison in Fig. 2b for surface-code memory with delayed-erasure handling and maximum-likelihood decoding. The public implementation instead computes a classical repetition-code analogue. Table 1 follows the links preserved in the frozen archive.

Table 1: A frozen cross-artifact assessment chain. Exact fields, source pointers and file-presence checks are supplied in `anc/audit_system_chain.json`.

Evidence layer	Observed record	Meaning for the source target
Source objects	Three Fig. 2b items: exact-time, no-SSR and SSR curves	The requested observable belongs to the surface-code method.
Implemented calculation	Repetition-code proxy; distance 5, loss probability 0.01, 200,000 shots	An executable mechanism model is available.
Local check	Proxy marked physically consistent; expected ordering holds	The declared proxy passes its local checks.
Assessment gate	Local physics status passed, but final eligibility false and gate exploratory-only	Passing those checks does not accept the source target.
Target disposition	Model mismatch; attempted, not reproduced	The surface-code circuit and required decoder remain missing.

This distinction is a useful system behavior: it preserves a runnable intermediate implementation while preventing its local checks from being presented as achievement of the paper's result. The earlier proxy record cites unavailable author materials; the diagnosis record instead identifies the absent surface-code and decoder implementation as a current system limitation. The public history contains only one version of the detailed state, so it does not establish who detected this mismatch, a timed review intervention, or an autonomous repair. Moreover, a detailed validation check and execution-attestation file cited by the final record are absent from the public snapshot. The evidence supports the recorded separation of proxy validity and target acceptance, not independent verification of the claimed isolated run.

3.4 Evidence used to assess capability

The examples below are drawn from public files at RunThePaper revision 43a64cd. We inspect the recorded implementations, reanalyze selected stored arrays and regenerate numerical figures. The accompanying analysis reads frozen Git objects and records input hashes; it does not execute the original scientific solvers. New work in this report consists of documentary linkage and consistency checks, figure regeneration, and selected arithmetic checks on stored values. Archived scientific labels remain historical observations rather than new verdicts. Historical execution and diagnostic claims are identified as such. The cases were selected for contrasting numerical tasks and inspectable outputs, so they establish concrete examples of implemented capability rather than an average success rate.

4 Demonstrated scientific capabilities

The following cases connect each capability to a scientific requirement, an implemented method and observable evidence. They also identify the remaining case-specific boundary needed to interpret that evidence.

4.1 Stable resource estimation for Hamiltonian simulation

The original target of PRAgent's qDRIFT case is the comparison of formula-based Hamiltonian-simulation resource bounds: it evaluates how Hamiltonian-simulation resource bounds depend on simulation time and an error criterion [6]. Its inputs include Hamiltonian summary parameters and the error-bound expressions associated with different compilation methods. The scientific output is an integer circuit-resource bound, which determines the numerical requirements of the implementation.

The archived implementation evaluates qDRIFT and deterministic or randomized product-formula bounds. It computes multiplicative expressions in logarithmic form and handles competing terms using stable log-

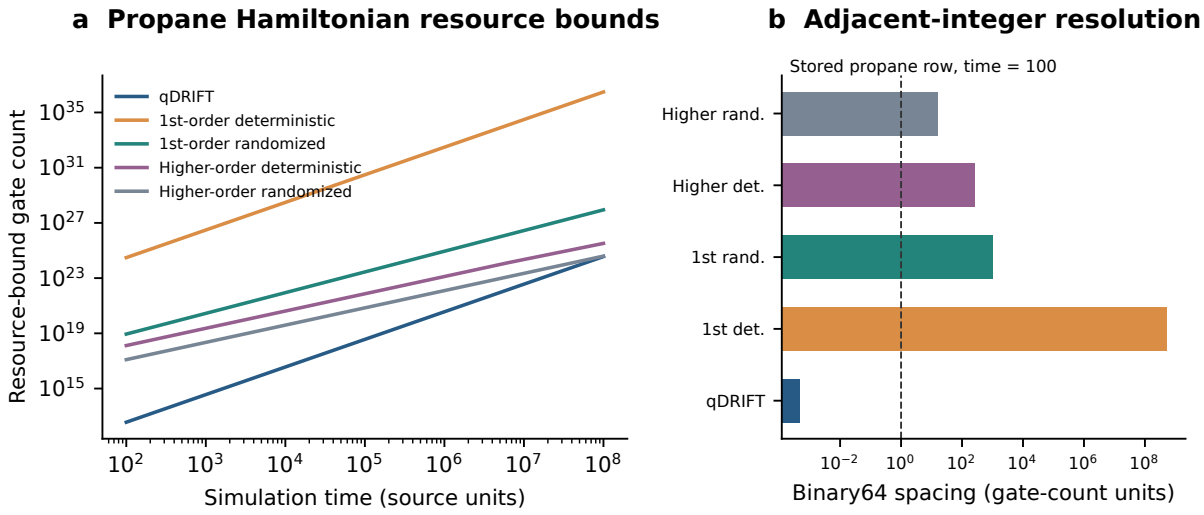


Figure 2: Resource-bound outputs and their precision requirement. **a**, Archived propane gate-count bounds for five compilation strategies. **b**, Binary64 spacing at each method's first stored propane count, at simulation time 100. The dashed line marks unit spacing; four counts cannot distinguish adjacent integers after conversion to binary64. Panel b is a new representation check on stored integers, not a rerun or independent certificate of the error-bound inequalities.

sum-exp calculations. These choices matter because the resource counts span many orders of magnitude. A naive evaluation can overflow even when the final comparison is meaningful.

A second issue arises when the output is the smallest integer satisfying a bound. At sufficiently large counts, adjacent integers have identical floating-point representations. The case implementation therefore uses a floating-point bracket as a starting point and refines the boundary using 60-digit decimal logarithms. The relevant question is whether the bound holds at N and fails at $N-1$, not merely whether a continuous estimate looks plausible. This is an example of a numerical-analysis requirement that emerges from the scientific output being requested.

The public tables contain 366 rows for time-dependent resource estimates and 366 rows for phase-estimation estimates. Figure 2a redraws the propane resource curves. Figure 2b tests the numerical representation required by the first stored propane row, at simulation time 100. Its deterministic first-order count is 3,056,542,447,900,034,673,748,014: the spacing between neighboring binary64 values at that magnitude is 536,870,912. Converting that integer and its predecessor to binary64 gives the same value. Four of the five methods in this row exhibit this adjacent-integer collapse. This direct check of the stored integers establishes why binary64 alone cannot resolve the requested integer boundary; it does not evaluate the physical error bound.

The code implements the boundary condition

$$\log E(N) \le \log \epsilon, \qquad N = 1 \text{ or } \log E(N-1) > \log \epsilon,$$

using decimal refinement after a floating-point estimate. An archived target-check record marks the minimal-integer checks passed for three methods across three molecules. The record supplies Boolean outcomes, not signed bound margins for every table row. Thus the public evidence supports a precision-aware implementation and recorded boundary checks; a fully independent certificate would require reevaluating both inequalities at the declared precision and conditions. We do not claim that additional verification here.

A separate printed-value discrepancy and missing independent review remain open. The capability demonstrated here is the implemented treatment of a specific numerical obstacle, with inspectable output integers and check records.

4.2 Interacting quantum walks in exact particle sectors

The source target in PRAgent's quantum-walk case is the density and correlation structure of interacting excitations. The public code implements the coherent theoretical model associated with the

superconducting-processor experiment of Yan and colleagues [7]. A Bose-Hubbard Hamiltonian provides the case formulation,

$$H = \sum_{j=1}^{L-1} J_j(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j) + \sum_{j=1}^{L} \frac{U_j}{2} n_j(n_j - 1) + \sum_{j=1}^{L} h_j n_j.$$

The implementation retains source-linked coupling and interaction parameters and works in fixed-particle-number sectors. The one-excitation calculation uses an 11-site sector for the corresponding source setup. The two-excitation calculation uses 12 sites, with 78 bosonic occupation states and 66 states in the hard-core limit. These representations implement the stated theoretical sectors; they do not reproduce the experimental hardware or its raw measurements.

The computational structure makes the physical model inspectable. The basis is enumerated as occupation tuples, and hopping matrix elements carry the bosonic factors $\sqrt{n_j(n_{j+1}+1)}$. A Hermitian eigendecomposition is reused to evaluate time evolution on the requested grid. Observable functions then compute site density and the two-particle correlator,

$$G_{ij} = \langle a_i^\dagger a_j^\dagger a_i a_j \rangle,$$

including the diagonal convention $G_{ii} = \langle n_i(n_i - 1) \rangle$. The calibrated interaction, free-boson and hard-core cases share a transparent comparison structure.

Figure 3a displays the stored one-excitation density for a central launch. The correlation matrices in Fig. 3b–d show the calibrated interaction and its two control models for the same two-excitation launch and displayed time. The shared solver and observable definitions allow a controlled comparison of how the interaction model changes the spatial correlation pattern.

We independently recomputed the density-sum diagnostic from the stored CSV values. Across 1,503 one-excitation state-time slices, the maximum deviation from one is 1.78×10^{-15}. Across 3,006 two-excitation slices, including the recorded model and launch combinations, the maximum deviation from two is 2.66×10^{-15}. These residuals establish internal conservation consistency. In a normalized fixed-number sector, the density sum equals the particle number by construction: incorrect couplings or an incorrect comparison time can still pass this test. It therefore supplies no quantitative measure of agreement with the source density or correlation curves.

The supported capability is the implemented connection between physical structure and computation: the conserved sector determines the basis, one propagation supplies several observables, and explicit limits provide controls. The public comparison still has an unresolved source-time discrepancy and missing author realization-level parameters. Until those conditions and direct observable errors are checked, this chain establishes model implementation and internal consistency rather than source-result reproduction.

4.3 Quantum discord: separating ensembles, references and logical claims

PRAgent's discord case demonstrates ensemble-based computation and interpretation of a numerical discrepancy. It implements the DQC1 correlation problem considered by Datta, Shaji and Caves [8]. For a unitary U acting on a register of dimension d and control polarization α, the case constructs

$$\rho = \frac{1}{2d} \begin{pmatrix} I_d & \alpha U^\dagger \\ \alpha U & I_d \end{pmatrix}.$$

It computes entropic quantities and measurement-dependent conditional entropies using explicit numerical representations. The source question motivates comparing the resulting discord with an analytical reference, while the implementation must also specify the unitary ensemble and optimization procedure.

The public numerical-method record describes two separately generated ensembles of 500 matrices in $U(32)$: a QR-based Haar construction and a depth-32 local brickwork construction. The latter is explicitly identified as an independent probe because the original finite pseudorandom generator is not published. Each curve is stored on a 41-point polarization grid, with mean, standard deviation, standard error and interval fields. This representation keeps sampling variation visible instead of presenting an individual realization as a universal result.

Figure 4 shows the archived means and their deviations from the analytical reference. At $\alpha = 1$, the stored QR-Haar mean is 0.537735 bits, the brickwork mean is 0.537483 bits and the reference is 0.557305 bits. Their corresponding differences from the reference are approximately -0.01957 and -0.01982 bits. The

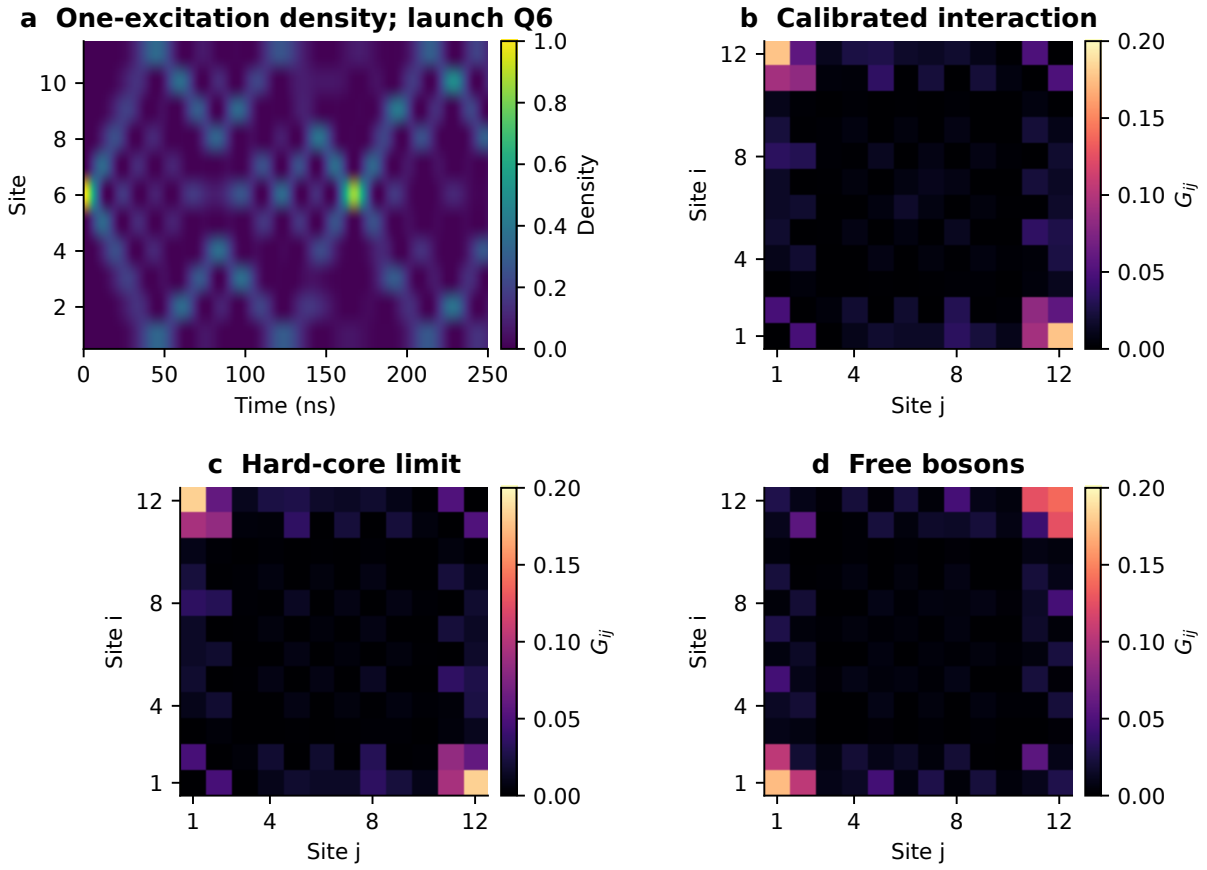


Figure 3: Quantum-walk outputs regenerated from stored arrays. **a**, One-excitation density after launch at site Q6 in the 11-site setup. **b–d**, Two-excitation correlators at 55.5 ns for launch Q6+Q7 in the 12-site setup, using calibrated interactions, the hard-core limit and free bosons. The correlation panels share a color scale. These are generated model outputs, not experimental measurements; the displayed time is taken from the archived table and is not an adjudication of the case's unresolved source-time issue.

displayed bands use the case's stored pointwise 95% interval endpoints. They describe ensemble uncertainty under the recorded procedure, not simultaneous confidence bands or uncertainty in the analytical expression.

Both generators feed the same entropy and measurement-optimization implementation. The archived production configuration specifies 129 projective axes for this shared routine. Agreement between the two ensemble means is therefore not a comparison of independent discord implementations. Their narrow intervals describe sampling variability and cannot exclude shared discretization, optimization or formula errors. The observed discrepancy is preserved in the archived outputs, but its cause remains unresolved; finite-size effects and shared numerical bias require discriminating checks before a physical or source-paper conclusion follows. The demonstrated capability is producing and exposing this comparison with its sampling conditions, not having resolved the discrepancy.

The code also makes a useful logical distinction in its treatment of entanglement. For the control-register split, it constructs a convex product-state decomposition from the eigenbasis of U. For other partitions, one-sided tests such as negativity are recorded as detection criteria. Failure to detect entanglement is not used as a separability proof. This illustrates how the interpretation of a numerical check can be as important as its implementation.

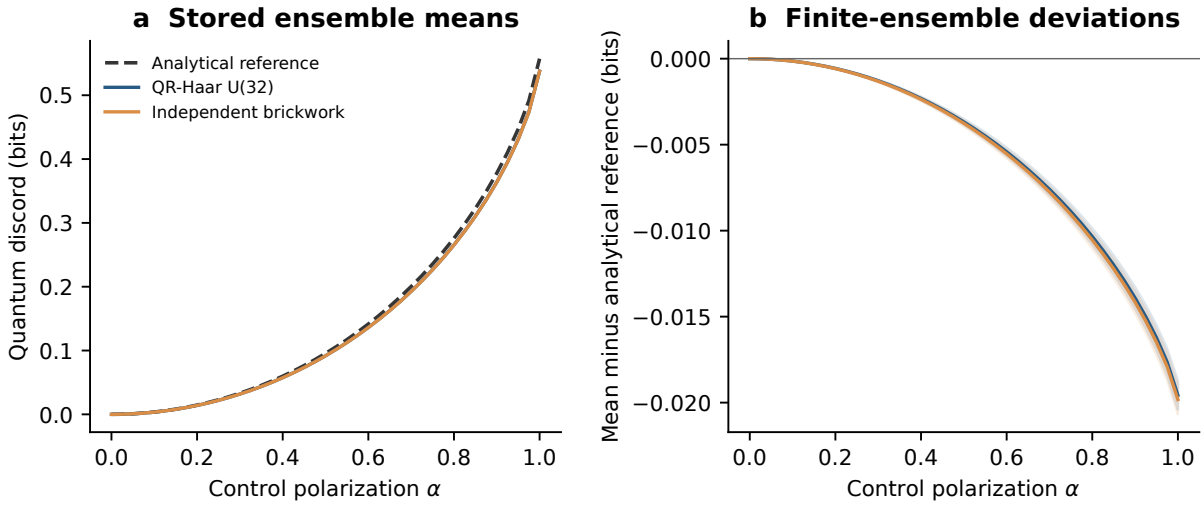


Figure 4: Quantum-discord outputs redrawn from the archived DQC1 case. **a**, The analytical reference and means of the QR-Haar and separately generated brickwork ensembles, evaluated through shared numerical routines. **b**, Mean-minus-reference residuals expose deviations that are less visible in the main curves. Shading uses the stored pointwise interval endpoints. Ensemble specifications originate in the public method record; these are generated case outputs rather than recovered author samples.

5 Evidence across the 100-paper project

5.1 Selection and source-result linkage

The analysis retains one fixed roster of 100 paper identities. It is a retrospective, purposive collection rather than a preregistered benchmark. The selection history contains 103 identities: three archived cases were replaced after evidence limitations were identified, and three low-compute studies were selected for planned work. The excluded cases lacked established comparisons for NiO arrays, source-condition rheology, or a 30-material comparison; all six identities and the recorded reasons are preserved in `anc/selection_provenance.json`. This outcome-informed selection limits population-level performance claims. The three planned studies remain unmeasured in this snapshot; they are included in the same roster without imputed scores.

The public extraction contains code for 97 cohort members and data for 96. To separate file availability from scientific support, we audit the linkage of archived result objects to source references, execution targets, declared parameter matches, reference types and cited public files. The inventory has 3,241 object records from 95 papers. Two further cases have assessment records but no object inventory; three have no measurements. Objects vary in granularity, including curves and result components, so their count is not a common scientific denominator.

The audit identifies 950 records whose mapped targets all declare paper-exact conditions and a numerical reference; 1,176 with a proxy, subset or reduced-scale comparison; 675 with unspecified or other conditions; 423 with figure or feature references; and 17 with mixed or insufficient comparison metadata. These are descriptions of archived evidence scope, not newly certified outcomes. In particular, the first category does not establish that the declared conditions or numerical comparator are correct. The accompanying object ledger preserves those declarations, source locations, gaps and cited-file availability so that a later scientific review can examine them individually.

5.2 What the historical percentages measure

Figure 5a–b reconstructs an archived table dated 27 August 2026. Across 97 measured papers it contains 1,399 numerical claims, 3,667 distinct checks and 16,376 claim–check associations. Fifteen non-numerical claims are excluded. Define

$$h_c = \frac{\#\{\text{checks associated with } c \text{ and labelled reproduced}\}}{\#\{\text{checks associated with } c\}}.$$

Associations include supporting checks, and the same check can contribute to several claims. The mean of h_c over claims is 40.9%; averaging first within each paper and then over measured papers gives 57.8%. These are archived check-label fractions. They do not measure the fraction of source results reproduced, the extent of agreement with source observables, or whole-paper completion.

The metric depends on how a check list is partitioned: duplicating one positive check can change h_c without changing a calculation or its agreement with the paper. The ancillary sensitivity table demonstrates this on mixed-label claims. Restricting associations to the primary claim would still not establish direct scientific support. We retain the historical fractions to describe the archive, and use the source-result ledger and detailed examples to identify what evidence each calculation actually supplies.

5.3 Joint states and conflicting records

The completion summaries mark numerical scope complete for 83 papers, science passed for 34, and independent review passed for 26. A separate historical set of 26 papers has $h_c = 1$ for every measured claim. Table 2 shows that these two sets of 26 are disjoint. Furthermore, only one paper has both summary-level science and review marked passed. Marginal counts therefore cannot be interpreted as successive stages of a successful reproduction funnel. Review status is also distinct from the scientific conclusion: case `1803.01876` has review passed, science failed and paper assessment inconclusive in both records. Thus a passed review label cannot itself be read as a positive reproduction certificate.

Table 2: Two separate contingency tables share the review columns; the first two rows and last two rows each sum to 97. Three unavailable cases remain outside each comparison. Case identities are supplied in the joint-state ledger.

Condition among 97 available cases	Review passed in summary	Review not passed in summary
Historical check-label fraction 100%	0	26
Historical check-label fraction below 100%	26	45
Science passed in summary	1	33
Science not passed in summary	25	38

Even within the same frozen Git revision, summary and detailed records disagree for six science states, six review states, seven execution states and one numerical-scope state. Additional conflicts concern assessment, provenance and evidence dimensions. We do not choose a preferred label without reconciliation. Figure 5c instead shows agreement, conflict and missingness explicitly: the two records agree on a positive scope state for 83 papers, a positive science state for 32 and a positive review state for 26. Agreement is consistency between records, not independent confirmation of their scientific validity.

Neither record set marks any available case complete. Among the 26 papers with all associated checks historically labelled reproduced, the detailed records show 20 reviews missing, three stale and three passed. They also record 22 attested executions and four missing attestations. Other overlapping blockers include seven cases with missing parameter provenance and five with targets not declared paper-exact. The three review-passed cases retain mixed or paper-error-candidate assessments and other blockers. This explains why local labels cannot establish completion; it does not reduce all incompleteness to a lack of compute or a single isolation-backend issue. Nor does a missing review prove that an otherwise unexamined scientific result is wrong.

6 Diagnostic capability and observed obstacles

PRAgent's diagnostic records connect an observed obstacle to a discriminating test or a missing scientific requirement. Source ambiguity, incomplete implementation, execution resources and experimental apparatus provide four directions for investigation. The three examples below illustrate this process without estimating cause prevalence. The consistency audit above preserves conflicts before diagnostic interpretation. Source pointers and recorded confidence for the examples are supplied in `anc/obstacle_examples.json`.

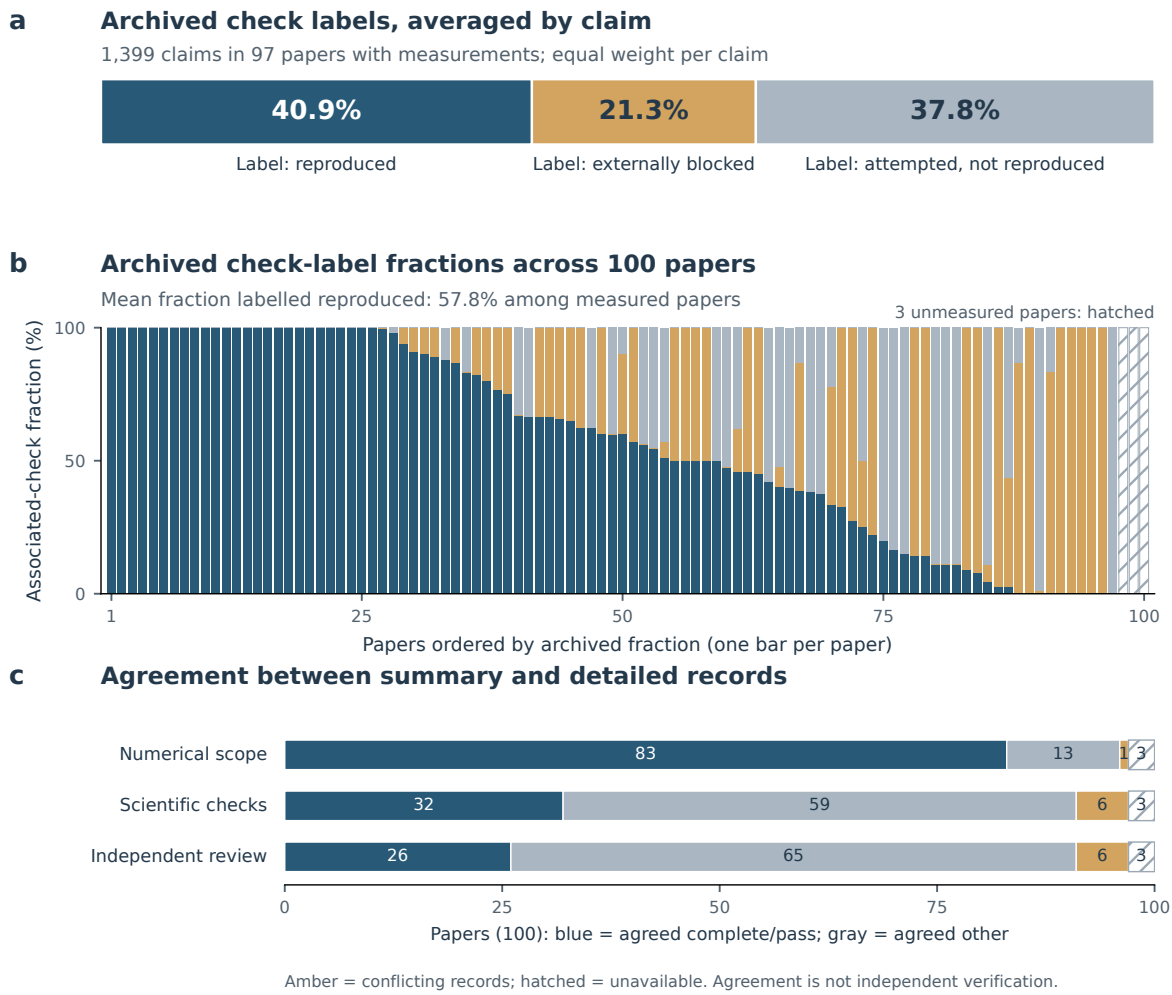


Figure 5: Historical check labels and assessment-record consistency for the fixed 100-paper project. **a**, Fractions of associated checks carrying each archived label, averaged over 1,399 claims. **b**, The same fractions by paper; hatched bars retain three unmeasured papers. The 57.8% mean weights measured papers equally. Supporting and shared checks contribute to both panels, so neither is a scientific reproduction rate. **c**, Agreement between completion summaries and detailed states, with conflicts and unavailable records shown separately and a fixed denominator of 100. Agreed positive means complete scope or passed science/review in both records; it is not a new verification verdict.

Testing source identifiability. For Fig. 2 of Phys. Rev. Lett. 124, 113601 (target T001), the archived diagnosis identifies unspecified finite-chain boundaries, site-index origin and localization conventions. The recorded source audit and identifiability test find that publication-compatible choices produce different energy and inverse-participation-ratio arrays and localized-state counts. This supports a limitation on identifying one exact numerical target from the disclosed inputs. Resolving it requires obtaining the missing conventions, fixing them before comparison and rerunning the calculation. Pending gates remain in the record; the diagnosis does not establish a false physical claim or independently exclude implementation error.

Checking implementation scope. The surface-code example in Table 1 distinguishes an incomplete scientific method from a missing author artifact. A local proxy can be tested and executable while failing the requested model. The required next capability is a surface-code circuit and lifecycle-aware decoder, followed by validation on tractable instances and evaluation of the source observable. Additional hardware alone would not supply that missing method.

Relating execution requirements to measurements. Case `1709.03478` (T002) records reduced-scale tests and one full-size stationary block, while the complete production and convergence campaign remains unexecuted. Its diagnosis reports 2,784 CPU tasks and 27.004229 seconds for the measured block, giving a 20.88 CPU-hour baseline before aggregation. Finite-time tasks may take longer; the available A100 has no validated CUDA implementation. This provides an execution estimate for the current method, without

separating resource allocation from scheduling or backend choices. Follow-up requires an allocated CPU campaign or a validated alternative backend, then convergence checks.

These are archived diagnostic examples, not new independent causal verdicts. Missing data or supplementary material does not establish missing experimental apparatus; that requires checking the physical operations and device access. Evidence availability also limits diagnosis: in `2506.06669`, a later pulse-definition explanation cites a detailed test file absent from the public snapshot. That explanation still needs inspectable support. Retaining such limits lets the workflow specify what evidence is needed next.

7 Capability boundaries and next evaluation steps

The cases demonstrate different components of PRAgent's scientific work: implementing the relevant mathematical object, handling numerical constraints, checking physical structure and retaining evidence for further investigation. The implementations are reusable beyond the original interaction. A particle-sector solver, precision-aware bound evaluator or ensemble-analysis routine can support a new calculation when its model assumptions remain appropriate.

These capabilities were exercised within a human-guided system. Historical model versions, interaction budgets and interventions are not uniformly recorded, so the evidence does not isolate an autonomous model's contribution or measure the gain over a human-only workflow. The fixed cohort is purposively selected, and the detailed examples were chosen with knowledge of their outputs. A controlled evaluation would need recorded intervention and resource budgets, common acceptance criteria and a comparison condition.

The report adds documentary re-adjudication of evidence scope and consistency, plus arithmetic checks on selected stored arrays. It does not independently revalidate every source comparison, execution or causal diagnosis. A cited-file presence check establishes availability only; it does not certify the file's scientific content. Missing public files limit external verification even where a private run may have existed. Later production-backend readiness is a separate deployment issue and cannot be used to explain all archived completion states.

The next evaluation should test whether these implemented capabilities transfer to new papers under a recorded protocol. Useful measures include agreement with independently checked observables, correct handling of model limits, the validity of diagnostic follow-up and the human effort needed to complete a task. Source clarification, missing method implementations and full-scale execution remain concrete development priorities identified by the cases.

8 Conclusion

PRAgent's public cases provide concrete implementations of precision-sensitive resource evaluation, conserved-sector quantum dynamics and finite-ensemble discord analysis. The surface-code example adds a system-level result: the assessment records retain a passing proxy calculation while leaving the original target unaccepted. These examples make the system's scientific representation, numerical methods and acceptance distinctions inspectable. The 100-paper audit also identifies where a broader capability claim would exceed the evidence: auxiliary check labels are not source-result success rates, review and science states often do not overlap, and some records conflict. The resulting source-object ledger makes the next evaluation concrete: verify the declared comparison conditions and numerical references, reconcile the assessment records, and test new papers with recorded human interventions and resource budgets.

Data, code and reproducibility

The public case archive is available at RunThePaper, revision 43a64cd. The three implementations are identified by case IDs `1811.08017`, `10.1126-science.aaw1611` and `0709.0548`. The report package includes input hashes, numerical summaries, a unified cohort roster, source-object and state-conflict ledgers, joint-state tables and the concrete proxy assessment chain. The accompanying analysis scripts read frozen public Git objects and reconstruct all new numerical figures without executing case code; the workflow schematic is retained as a conceptual diagram. The private production environment is not included in this release.

AI assistance and manuscript responsibility

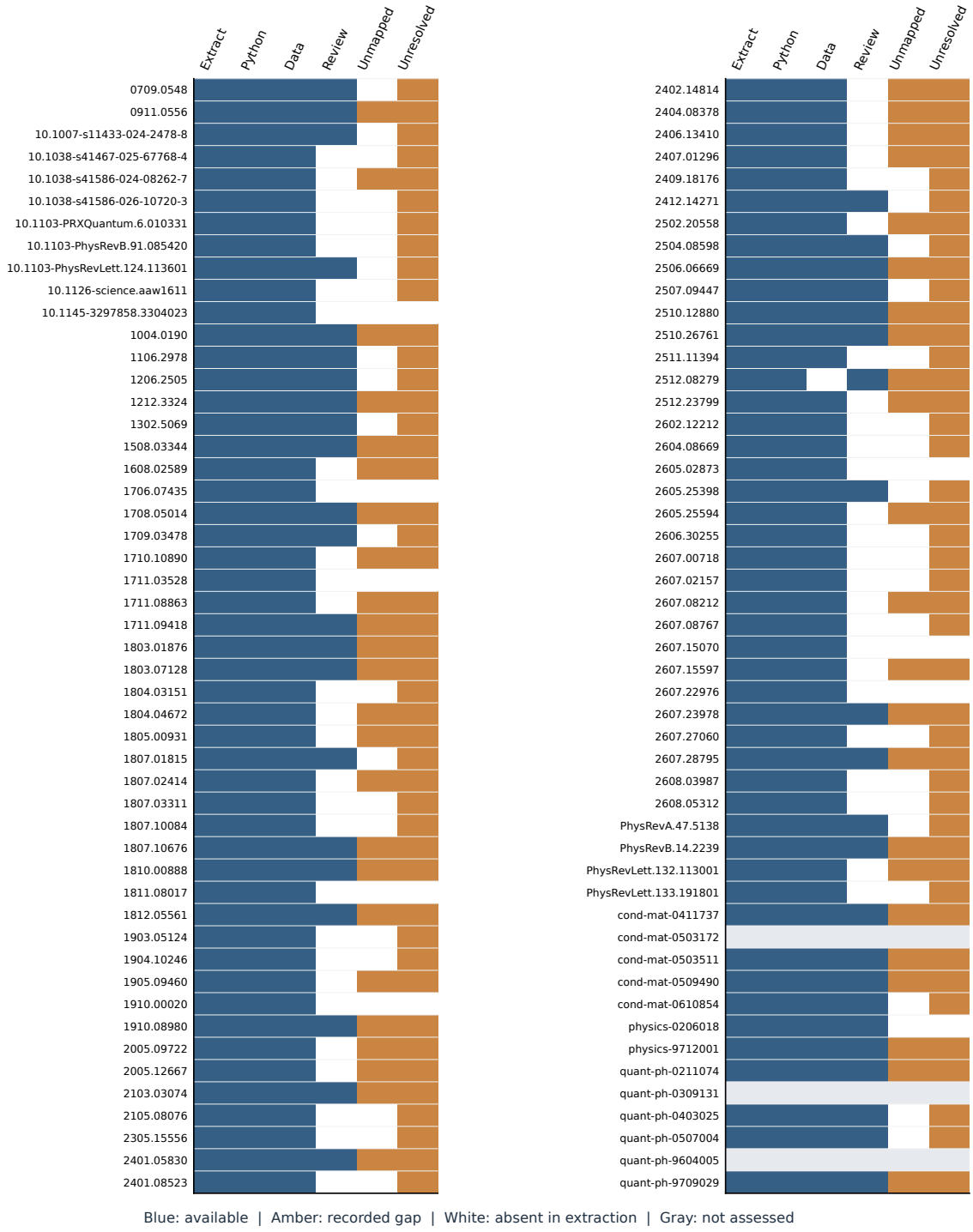
AI-assisted tools were used in software development, manuscript preparation and the visual design of the workflow schematic. Numerical figures were plotted from the archived arrays. The report does not attribute authorship to an AI system or claim a uniformly autonomous experimental campaign. The human authors are responsible for the interpretation of results, citations and final submitted text.

References

1. Sandve, G. K., Nekrutenko, A., Taylor, J. and Hovig, E. Ten Simple Rules for Reproducible Computational Research. *PLoS Computational Biology* **9**, e1003285 (2013). doi:10.1371/journal.pcbi.1003285.
2. Pineau, J. et al. Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program). *Journal of Machine Learning Research* **22**(164), 1–20 (2021). jmlr.org/papers/v22/20-303.html.
3. Starace, G. et al. PaperBench: Evaluating AI's Ability to Replicate AI Research. *arXiv* **2504.01848** (2025). arxiv.org/abs/2504.01848.
4. Siegel, Z. S., Kapoor, S., Nagdir, N., Stroebl, B. and Narayanan, A. CORE-Bench: Fostering the Credibility of Published Research Through a Computational Reproducibility Agent Benchmark. *arXiv* **2409.11363** (2024). arxiv.org/abs/2409.11363.
5. Chan, J. S. et al. MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering. *arXiv* **2410.07095** (2024). arxiv.org/abs/2410.07095.
6. Campbell, E. A random compiler for fast Hamiltonian simulation. *Physical Review Letters* **123**, 070503 (2019). doi:10.1103/PhysRevLett.123.070503. Source preprint: arXiv:1811.08017v2.
7. Yan, Z. et al. Strongly correlated quantum walks with a 12-qubit superconducting processor. *Science* **364**, 753–756 (2019). doi:10.1126/science.aaw1611.
8. Datta, A., Shaji, A. and Caves, C. M. Quantum Discord and the Power of One Qubit. *Physical Review Letters* **100**, 050502 (2008). doi:10.1103/PhysRevLett.100.050502. Source preprint: arXiv:0709.0548.

Supplement: unified project roster

The project retains all 100 identities in one roster. The matrix below records evidence availability and mapping gaps, without defining origin-based comparison groups. Blue indicates available extracted material, amber a recorded gap, white absence within an available extraction and gray a measurement not assessed. These indicators describe public evidence organization, not scientific-success grades. Full titles and the selection provenance are supplied with the accompanying data.



Supplementary Fig. S1. Evidence availability for the fixed 100-paper project.